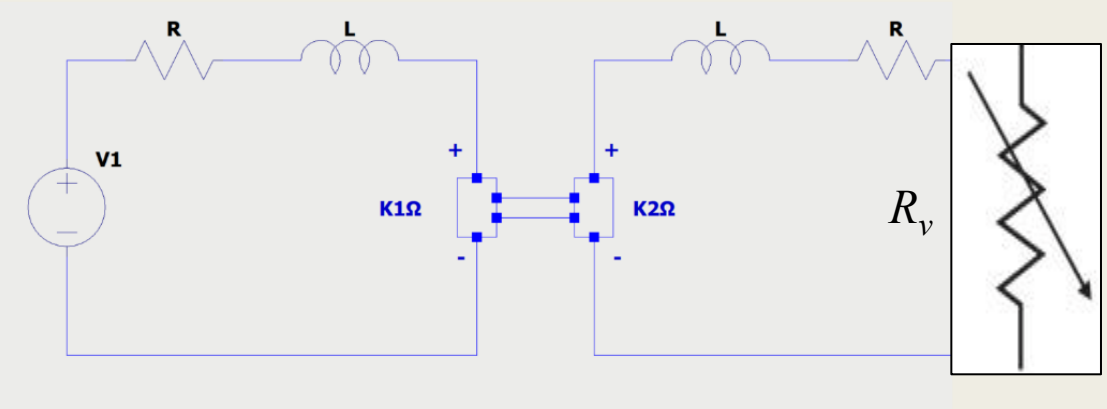
LABORATORY

Robust Control and Multi-variable Systems

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Deriving the model



Two coupled Dc motors

- Electric model
- Mechanical model
- Define input, output $\rightarrow V_1(s), \Omega(s)$
- Find an operating point for some fixed (middle) value R_v $\rightarrow V_1^*, \Omega^*$
- Calculate the model in deviation variables $\rightarrow \delta V_1(s), \delta\Omega(s)$
- Select maximum deviations and scale the model $\rightarrow y(s) = G(s)u$
- Simplify the model (1st order transfer function $G(s)$) and **identify** the scaled model

  (steady state gain and time constant) $\rightarrow G(s) \cong \frac{K_u}{\tau_u s + 1}$

Deriving the model

Identification

- Keeping R_v fixed measure $y(t)$ for $u(t) = \bar{u} \in [-1,1]$ until the system is settled.
- For the various inputs $\bar{u} \in [-1,1]$ check the steady state value of $y(t)$ and use least squares to find the gain K_u and find the time constant similarly by the raise time.
- Repeat for $R_v = 0$ and $R_v = max$.

Validation

- Perform (for all values of R_v) a frequency response analysis (gain, phase) by adopting $u(t) = 0.3\sin(\omega t)$ for various relatively small frequencies.
- Plot the frequency response of the identified model and compare with your measurements.

Designing controllers

Loop shaping design

- Use the inverse-based controller design for command tracking.
- Use the loop shaping for disturbance rejection (P, PI, PID).
- Use the 2-DoFs design to meet both control objectives.
- Use H∞-mixed sensitivity design.
- Check how your closed loop system behaves when you exert torque on the axis by your finger.

Performance specifications

- Select time and frequency specifications and include them in your design.

Implementation

- First perform simulations to validate your design and then implement the controllers.
- Discretize the transfer functions that have been designed in the previous steps using the sampling time. Alternatively, use Euler discretization of the state space model of the controller.

Modeling the uncertainty & robust control design

- Use the bode diagram (phase and amplitude) you have calculated to estimate the uncertainty with respect to the nominal first order model you have identified in a previous step. Select various values of the resistor R_v around the nominal value.
- Check based on the extracted uncertainty model the Robust Stability and Robust Performance properties of all aforementioned control schemes [use the same weighting functions with the previous control design step].
- Design a robust controller with the same specifications that enforces both robust stability and robust performance.

Multivariable systems

- Study the papers A.pdf and B.pdf in the Documents folder of eclass that consider a process with 4 tanks.
- Model the process.
- Construct the model in the lab. Four bottles will be employed, attached to force cells to measure the weight and thus the water level. Two water pumps will also act as the actuators.
- Identify the parameters of the model. Apply various voltages at the pumps and measure the volume of the water. Then without activating the pumps measure how fast the water outlets the tanks.
- Perform analysis on the model, design a controller to regulate the water level and simulate in MATLAB.